Forage Crop Classification in Heterogeneous Environments Using UAV, Machine Learning, and RGB-Multispectral Fusion

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***Abstract*—The integration of red–green–blue (RGB) and multispectral imagery (MSI) captured by uncrewed aerial vehicle (UAV) combined with machine learning (ML) techniques presents a promising approach for the classification of forage crops in heterogeneous agricultural systems. This study evaluated the predictive performance of 19 ML algorithms using RGB, MSI, and combined RGB + MSI data. XGBoost consistently demonstrated the best balance between accuracy, generalization, and computational efficiency, achieving high performance across multiple evaluation metrics, particularly when using the fused dataset. Although random forest (RF) showed competitive accuracy, it was outperformed in terms of generalization and processing time. Hyperparameter tuning and model blending failed to improve predictive performance and, in some cases, led to performance declines, especially in the classification of phenotypically similar classes. The analysis revealed that model errors were concentrated among crops with similar spectral and structural characteristics, such as Mombaça grass, Tanzania grass, and forage cactus clones, underscoring challenges in distinguishing between such groups even at high spatial resolutions. These findings emphasize the importance of data quality, feature selection, and model interpretability in remote sensing-based classification tasks. The results have practical implications for precision agriculture, supporting more accurate crop monitoring through UAV-based remote sensing. Future research should incorporate variable selection methods, agronomic data, and deep learning (DL) approaches to enhance classification in complex agricultural environments.**

***Index Terms*—Crop classification, ensemble learning, model interpretability, precision agriculture, spectral data fusion.**

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I. Introduction

PRECISION agriculture has emerged as a key strategy for optimizing natural resource use and promoting sustainable management in agricultural systems, particularly in the context of global population growth and the increasing demand for food, grains, fibers, and fuels [1]. The pursuit of increased agricultural productivity without compromising ecosystem integrity highlights the importance of innovative methods for crop identification and monitoring [1], [2]. Recent advances in remote sensing have provided high-resolution data essential for analyzing variations in heterogeneous agricultural systems [3], [4].

Numerous studies have focused on the crop classification process in agricultural environments, including the use of satellite imagery [1], [5], [6], [7], [8]. However, the complexity of agricultural systems (intensified by the diversity of crop types) demands methodologies capable of addressing such heterogeneity [9], [10]. Consequently, the low spatial and temporal resolution of orbital sensors, along with the interference of factors such as cloud cover and mixed pixels, can undermine classification accuracy [11], [12]. Crop identification based on imagery acquired by uncrewed aerial vehicles (UAVs) has therefore emerged as a promising alternative for improving agricultural management [13], [14].

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The use of UAVs equipped with various sensors (particularly the red–green–blue (RGB) and multispectral cameras) has revolutionized crop monitoring by offering advantages such as low cost, high accuracy, and greater planning flexibility [13], [15]. These sensors enable the collection of more detailed crop information, facilitating both intra and interclass differentiation. The type of sensor directly influences the level of detail that can be obtained as well as the associated acquisition and processing costs. For instance, RGB imagery is affordable and easy to process but offers limited spectral capabilities [16]. In contrast, multispectral sensors provide a balance between cost and information richness, allowing for more detailed vegetation analysis [17]. Hyperspectral sensors, in turn, capture a broad range of wavelengths, delivering highly precise data, though at higher costs and with greater operational complexity [18].

In agricultural applications, the use of UAVs for classification tasks has been explored in the identification of various crop types [2], [19], [20], [21], [22]. While the adoption of these technologies in forage crop systems remains limited, despite the critical role these species play in productive systems [23]. Forage crops are fundamental to livestock production, ensuring feed availability for herds and contributing to the economic stability of the sector, in addition to providing ecosystem services such as the restoration of degraded areas and carbon assimilation [24], [25], [26]. Thus, expanding the use of UAVs in this domain is important for improving the management and productivity of these systems. Machine learning (ML)-based methods, combined with the extraction of specific spectral indices, show potential for enhancing the accuracy of forage crop identification and mapping.

ML algorithms such as XGBoost and random forest (RF) have stood out for their ability to handle complex data and their efficiency in minimizing classification errors [19], [27], [28], [29]. Alzhanov and Nugumanova [19] recommended algorithms such as the extra trees classifier (ETC) and RF for classifying soybean, barley, and wheat, while Lee et al. [28] employed XGBoost for classifying objects such as rice, beans, fruit, infrastructure, corn, sugarcane, and water, achieving a better balance of accuracy. Model ensemble strategies, in which different models are combined into a single predictor, have also emerged as promising alternatives to enhance predictive capability, particularly in scenarios where data variability hinders the development of generalizable models [30], [31], [32], although the application of this technique in the context under discussion remains limited. Automated ML (AutoML) frameworks, such as PyCaret [33], have gained prominence for their ability to integrate model comparison, hyperparameter optimization, and ensemble generation into unified workflows, raising important discussions about the extent to which automation can be adapted to domain-specific while preserving interpretability and methodological rigor [34].

Furthermore, recent advances in deep learning (DL)-based data fusion frameworks, such as convolutional neural networks (CNNs) and vision transformers (ViTs), have demonstrated remarkable capabilities in processing multisensor data for crop classification through the automatic extraction of hierarchical spatial and spectral features [35], [36], [37], [38]. These adaptive joint learning approaches enable the integration of multiscale information, often outperforming traditional ML methods in scenarios involving massive datasets [39]. Nevertheless, the application of DL in precision agriculture still faces challenges related to high computational demands, the need for extensive labeled datasets, and the limited interpretability of models when compared with traditional ML model architectures [35], [39].

Recent studies have pushed the boundaries of remote sensing by integrating high-dimensional data, such as hyperspectral imagery combined with light detection and ranging (LiDAR) for vegetation species classification in wetlands using adaptive ensemble learning stacking [40], or by the combination of satellite-derived synthetic aperture radar (SAR) imagery with multispectral data acquired from both satellite and UAV platforms, integrated through multiscale feature representations and processed via gradient boosting models for mangrove mapping [41], or by synergistically combining hyperspectral and SAR imagery through adaptive ensemble learning and 1-D-CNN architectures for vegetation parameter retrieval [42]. While these approaches offer superior accuracy in complex natural ecosystems, the high cost of sensors and the computational burden of processing such data pose significant barriers to their widespread adoption in precision livestock farming. Therefore, a critical knowledge gap persists regarding the optimization of classification workflows for phenotypically similar forage crops (e.g., cacti clones and C4 grasses) using more accessible and cost-effective sensors [23]. This study advances the field by systematically evaluating the marginal gains of RGB–MSI fusion at ultrahigh spatial resolution, identifying the limits of model complexity in a heterogeneous agricultural environment.

This study aims to address the following questions.

1) Does the fusion of RGB and multispectral data improve the classification of forage crops compared to the isolated use of each sensor?
2) Which ML algorithm, among a set of 20 models, offers the best balance between accuracy, generalization, and computational efficiency in a heterogeneous environment?
3) Can hyperparameter optimization strategies and model blending outperform the best individual model in a scenario of high interclass similarity?

II. Materials and Methods

A. Study Area

The study was conducted at the International Reference Center for Agrometeorological Studies on Cactus and Other Forage Crops, located at the Federal Rural University of Pernambuco/Academic Unit of Serra Talhada (UFRPE/UAST) in Serra Talhada, Pernambuco, Brazil (Fig. 1). The region is characterized by air temperatures ranging from 20.1 °C to 32.9 °C, an average annual rainfall of 642 mm, and an average evapotranspiration demand of 1800 mm per year.

The study area is characterized by a diversity of experiments and interactions between forage crops, such as different clones of cacti: [IPA: IPA Sertânia (*Nopalea cochenillifera*),

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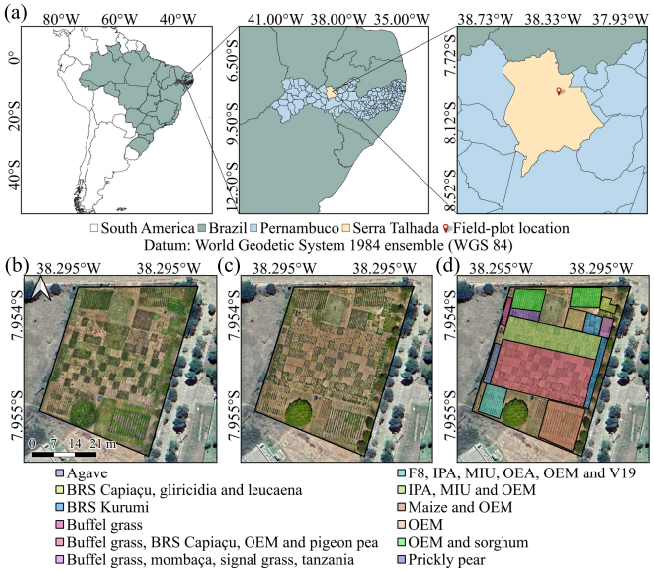


Fig. 1. Location and description of the study area. (a) Geographical location of the study area. (b) Orthomosaic from July 5, 2025. (c) Orthomosaic from December 9, 2025. (d) Description of the experimental area, where: clones of cacti: [IPA: IPA Sertânia (*Nopalea cochenillifera*), MIU: Miúda (*N. cochenillifera*) and OEM: Orelha de Elefante Mexicana (*Opuntia stricta*) and prickly pear (*Opuntia ficus-indica*)], agave (*Agave sisalana*), elephant grass (*Pennisetum purpureum* cv. BRS Kurumi and *Pennisetum purpureum* cv. BRS Capiaçu), buffel grass (*Cenchrus ciliaris* L.), guinea grass (*Megathyrsus maximus* cv. Mombaça and *Megathyrsus maximus* cv. Tanzania), signal grass [*Urochloa decumbens* (Stapf) R.D. Webster] (syn. *Brachiaria decumbens* Stapf), maize (*Zea mays* L.), sorghum [*Sorghum sudanense* (Piper) Stapf], gliricidia [*Gliricidia sepium* (Jacq.) Kunth ex Walp.], leucaena [*Leucaena leucocephala* (Lam.) de Wit] and pigeon pea [*Cajanus cajan* (L.) Huth].

MIU: Miúda (*N. cochenillifera*), OEM: "Orelha de Elefante Mexicana" (*Opuntia stricta*) and prickly pear (*Opuntia ficus-indica*)], agave (*Agave sisalana*), elephant grass (*Pennisetum purpureum* cv. BRS Kurumi and *Pennisetum purpureum* cv. BRS Capiaçu), buffel grass (*Cenchrus ciliaris* L.), guinea grass (*Megathyrsus maximus* cv. Mombaça and *Megathyrsus maximus* cv. Tanzania), signal grass [*Urochloa decumbens* (Stapf) R.D. Webster] (syn. *Brachiaria decumbens* Stapf), maize (*Zea mays* L.), sorghum [*Sorghum sudanense* (Piper) Stapf], gliricidia [*Gliricidia sepium* (Jacq.) Kunth ex Walp.], leucaena [*Leucaena leucocephala* (Lam.) de Wit] and pigeon pea [*Cajanus cajan* (L.) Huth].

B. Image Acquisition

Images were collected using a Mavic 3M UAV (DJI, Shenzhen, China) equipped with a 20-MP RGB sensor (4/3 CMOS) and four 5-MP multispectral sensors (1/2.8 CMOS) covering the green (560 $\pm$16 nm), red (650 $\pm$16 nm), red-edge (RE, 730 $\pm$16 nm), and near-infrared (NIR, 860 $\pm$26 nm) bands. The flight parameters were optimized to ensure high spatial resolution and minimize spectral mixing [43]. The flight plan was executed at an altitude of 30 m (yielding a GSD of 1.41 cm $\cdot$ pixel^{-1} to allow for the identification of fine leaf textures and pure pixel extraction), a speed of 2 m $\cdot$ s^{-1} (to prevent motion blur), and 80% front and side overlap (to ensure high-quality geometric reconstruction in dense forage canopies). Image coordinates were obtained via the GNSS onboard the UAV and were postprocessed using 20 ground control points (GCPs) fixed within the study area, georeferenced with the aid of a STONEX 850[a] device (Stonex, Paderno Dugnano, Italy). Image acquisition was scheduled between 11:00 A.M. and 12:00 P.M., during a period of consistent illumination and in the absence of clouds, to ensure higher image quality. Radiometric calibration was performed to convert raw digital numbers into surface reflectance values. This process was carried out by utilizing the downwelling light sensor (DLS) data and factory-calibrated parameters provided in the Mavic 3M metadata. All correction procedures and the generation of orthomosaics were carried out using Agisoft Metashape software (version 2.1.2, Agisoft LLC, St. Petersburg, Russia). During photogrammetric processing in Agisoft Metashape, all bands were automatically co-registered utilizing the internal camera calibration and sensor offset metadata provided by the Mavic 3M. Visual inspection of the resulting orthomosaics indicated no detectable spatial misalignment among spectral bands, and therefore no additional manual registration procedure was required.

TABLE I
FREQUENCY OF SAMPLES PER CLASS

Class	Frequency
Agave	27407
Signal grass	24162
BRS Kurumi	59849
Buffel grass	59932
BRS Capiaçu	51552
Gliricidia	30445
Pigeon pea	59999
IPA	59732
Leucaena	16790
Maize	59905
MIU	59980
Mombaça	27478
Mulching	59969
OEM	59321
Prickly pear	39583
Soil	60000
Sorghum	59873
Tanzania	18880

Where: clones of cacti: [IPA: IPA Sertânia (*Nopalea cochenillifera*), MIU: Miúda (*N. cochenillifera*) and OEM: 'Orelha de Elefante Mexicana' (*Opuntia stricta*) and prickly pear (*Opuntia ficus-indica*)], agave (*Agave sisalana*), elephant grass (*Pennisetum purpureum* cv. BRS Kurumi and *Pennisetum purpureum* cv. BRS Capiaçu), buffel grass (*Cenchrus ciliaris* L.), guinea grass (*Megathyrsus maximus* cv. Mombaça and *Megathyrsus maximus* cv. Tanzania), signal grass [*Urochloa decumbens* (Stapf) R.D. Webster] (syn. *Brachiaria decumbens* Stapf), maize (*Zea mays* L.), sorghum [*Sorghum sudanense* (Piper) Stapf], gliricidia [*Gliricidia sepium* (Jacq.) Kunth ex Walp.], leucaena [*Leucaena leucocephala* (Lam.) de Wit] and pigeon pea [*Cajanus cajan* (L.) Huth].

C. Data Extraction and Labeling

The dataset consisted of three image collections: RGB, MSI, and RGB + MSI. To ensure spatial resolution compatibility when merging RGB and MSI images, the multispectral images were resampled to an equivalent resolution using bilinear interpolation. Ground truth labels were generated through manual annotation of the orthomosaics in a Geographic Information System (QGIS v. 3.28). Georeferenced points for each class

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TABLE II
VEGETATION INDICES APPLIED IN THE STUDY

Sensor	Index Name	Abbr.	Equation	Ref.
MSI	Normalized Difference Vegetation Index	NDVI	$\frac{NIR-R}{NIR+R}$	[44]
	Normalized Difference Red Edge	NDRE	$\frac{NIR-RE}{NIR+RE}$	[45]
	Normalized NIR Index	NNIR	$\frac{NIR}{NIR+RE+G}$	[46]
	Ratio Vegetation Index	RVI	$\frac{NIR}{R}$	[47]
	Green Normalized Difference Vegetation Index	GNDVI	$\frac{NIR-G}{NIR+G}$	[48]
	Modified Chlorophyll Absorption in Reflectance Index	MCARI	$((RE-R)-0.2\times(RE-G))\times\frac{RE}{R}$	[49]
	MERIS Total Chlorophyll Index	MTCI	$\frac{NIR-RE}{RE-R}$	[50]
	Triangular Vegetation Index	TVI	$0.5\times(120\times(NIR-G)-200\times(R-G))$	[51]
	Spectral Feature Depth Vegetation Index	SFDVI	$\frac{NIR+G}{2}-\frac{R+RE}{2}$	[52]
	Soil Adjusted Vegetation Index	SAVI	$\frac{(NIR-R)\times(1+0.5)}{NIR+R+0.5}$	[53]
	Green Optimal Soil Adjusted Vegetation Index	GOSAVI	$\frac{(1+0.16)\times(NIR-G)}{NIR+G+0.16}$	[54]
	Normalized Green Red Difference Index	NGRDI_M	$\frac{G-R}{G+R}$	[55]
	Green-red Ratio Index	GRRI_M	$\frac{G}{R}$	[56]
	Optimized Soil Adjusted Vegetation Index-Green	OSAVIgreen_M	$\frac{1.5\times(G-R)}{(G+R)+0.16}$	[46]
	Enhanced Vegetation Index 2-Green	EVI2green_M	$\frac{2.5\times(G-R)}{G+2.4\times R+1}$	[46]
	Excess Red Vegetation Index	ExR_M	$1.4\times R-G$	[57]
RGB	Normalized Green Red Difference Index	NGRDI	$\frac{G-R}{G+R}$	[55]
	Green-red Ratio Index	GRRI	$\frac{G}{R}$	[56]
	Optimized Soil Adjusted Vegetation Index-Green	OSAVIgreen	$\frac{1.5\times(G-R)}{(G+R)+0.16}$	[46]
	Enhanced Vegetation Index 2-Green	EVI2green	$\frac{2.5\times(G-R)}{G+2.4\times R+1}$	[46]
	Excess Red Vegetation Index	ExR	$1.4\times R-G$	[57]
	Excess Green Vegetation Index	ExG	$2\times G-R-B$	[57]
	Excess Blue Vegetation Index	ExB	$1.4\times B-G$	[57]
	Normalized Difference Vegetation Index RGB	NDVIrgb	$\frac{(G+B)-R}{(G+B)+R}$	[46]
	Green Leaf Index	GLI	$\frac{2\times G-R-B}{2\times G+R+B}$	[58]
	Normalized Pigment Chlorophyll Ratio Index	NPCI	$\frac{R-B}{R+B}$	[59]
	Visible Atmospherically Resistant Index	VARI	$\frac{G-R}{G+R-B}$	[60]
	Woebbecke Index	WI	$\frac{G-B}{R-B}$	[57]
	Normalized Red	Rn	$\frac{R}{R+G+B}$	[56]
	Normalized Green	Gn	$\frac{G}{R+G+B}$	[56]
	Normalized Blue	Bn	$\frac{B}{R+G+B}$	[56]
	Color Intensity Index	INT	$\frac{R+G+B}{3}$	[61]
	Brightness Index	BI	$\sqrt{\frac{R^2+G^2+B^2}{3}}$	[62]
	Overall Hue Index	HUE	$\arctan\left(\frac{2\times(B-G-R)}{\sqrt{3}(G-R)}\right)$	[63]

Abbr.: Abbreviation, R: Red, B: Blue, G: Green, NIR: Near infrared (860 ± 26 nm), and RE: Red edge (730 ± 16 nm).

were delineated directly on the ultrahigh-resolution orthomosaics, avoiding mixed pixels from soil, shadows, or adjacent vegetation. Circular buffers of 5 cm^2 were then generated around each point to define the regions of interest (ROIs) used for pixel extraction. The distribution of samples per class is presented in Table I.

D. Vegetation Index

After extracting the bands by ROI, 34 vegetation indices were calculated (18 RGB and 16 MSI, Table II), adapting some indices that, by band equivalence, could also be calculated using MSI or RGB bands (Table II). The selection of vegetation indices aimed to represent a broad range of spectral transformations commonly used to characterize vegetation biophysical and biochemical properties. Indices derived from the multispectral sensor were selected because they incorporate the NIR and RE regions, which are strongly associated with chlorophyll concentration, canopy structure, and vegetation vigor [23]. In contrast, RGB-based indices were included to assess the discriminatory potential of visible-spectrum information, which is widely available in UAV-based monitoring systems [15]. Many of these indices emphasize differences among the red, green, and blue channels to enhance vegetation contrast and canopy detection. In addition, some indices were adapted between sensors when spectral equivalence allowed their formulation using both RGB and multispectral bands.

E. Modeling

1) Fit Models: We utilized PyCaret (version 3.3.2) [33] to automate model comparison and experimental configuration. The framework was selected due to its ability to provide a standardized and reproducible environment for benchmarking diverse ML algorithms, ranging from traditional linear models to state-of-the-art boosting ensembles, within a unified preprocessing and validation pipeline. During the preprocessing

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TABLE III
CLASSIFICATION MODELS LISTED IN PYCARET [33]

Model	Abbreviation	Reference
Ada Boost Classifier	AdaBoost	sklearn
CatBoost Classifier	CatBoost	catboost
Decision Tree Classifier	DT	sklearn
Dummy Classifier	Dummy	sklearn
Extra Trees Classifier	ETC	sklearn
Extreme Gradient Boosting	XGBoost	xgboost
Gaussian Process Classifier	GPC	sklearn
Gradient Boosting Classifier	GBC	sklearn
K Neighbors Classifier	KNN	sklearn
Light Gradient Boosting Machine	LightGBM	lightgbm
Linear Discriminant Analysis	LDA	sklearn
Logistic Regression	LR	sklearn
MLP Classifier	MLP	sklearn
Naive Bayes	NF	sklearn
Quadratic Discriminant Analysis	QDA	sklearn
Random Forest Classifier	RF	sklearn
Ridge Classifier	Ridge	sklearn
SVM - Linear Kernel	SVM-L	sklearn
SVM - Radial Kernel	SVM-R	sklearn

phase, we applied a z-score normalization process [see (1)] to standardize the scale of the predictor variables. This step was essential to ensure that the algorithms were not biased by the differing magnitudes of spectral bands and vegetation indices, thereby allowing a fair performance comparison across all models

$$z = \frac{x - \mu}{\sigma} \tag{1}$$

where z denotes the standardized value, x denotes the original observed value, μ denotes the mean of the feature, and σ denotes the standard deviation.

No feature selection procedure was applied to the dataset; all vegetation indices and raw spectral bands were retained throughout the entire modeling process. This was done to evaluate the algorithms' performance under high-dimensional conditions and to ensure that no potentially discriminative information was excluded. This approach also allowed for assessing the inherent ability of different algorithms to handle redundant or correlated predictors through their internal regularization and feature importance mechanisms. Following the identification of the best-performing model, its interpretability was assessed by calculating the mean absolute Shapley additive explanations (SHAPs) values [64]. This metric was used to quantify the global influence of each spectral feature on the model's final decision-making process.

To minimize the risk of overfitting, the dataset was partitioned into two independent sets: 70% for the training and model selection phase and 30% for the final performance evaluation (unseen testing set). The 70% partition was further subjected to a tenfold cross-validation procedure, where the data was internally and iteratively split into training and validation subsets. To prevent immediately adjacent pixels from being present in both partitions, the division process was performed according to the ROIs used for pixel extraction. In total, 19 models are available for classification in PyCaret (Table III), and the environment was configured to return the best-performing models evaluated during the training process.

2) Model Evaluation: The models were evaluated using the confusion matrix and a comprehensive set of derived metrics, including balanced accuracy [see (2)], precision [see (3)], recall [see (4)], Matthews correlation coefficient [MCC, (5)], and processing time. MCC was selected as the primary metric for model optimization and comparison, as it provides a comprehensive evaluation by considering all four quadrants of the confusion matrix, which makes it a dependable measure even when assessing results from imbalanced classes [65]. In this study, MCC and balanced accuracy are computed individually for each class using a one-vs-rest approach, where the class of interest is treated as the positive class and all others as negative. As shown in (1), it is defined as the average of sensitivity (true positive rate) and specificity (true negative rate), allowing for a balanced evaluation of performance even in the presence of class imbalance. This formulation was adopted for class-wise evaluation, particularly to highlight the model's behavior on minority classes and to ensure a more detailed understanding of classification performance across heterogeneous categories. Values of 1 for precision, recall, MCC, and balanced accuracy indicate perfect classification performance. The processing time was recorded in seconds, encompassing the full training and tenfold cross-validation cycle for each algorithm to provide a standardized metric of computational demand. All modeling and benchmarking procedures were performed on a workstation equipped with an Intel Core i7-10700KF CPU at 3.80 GHz, 32 GB of RAM, and an NVIDIA GeForce RTX 3060 GPU 12-GB GDDR6, running Windows 11 Pro version 25H2. The software environment was managed using Python 3.10. All models were executed under identical system priority settings, and times were recorded using the internal logging system of the PyCaret library

$$\text{Balanced Accuracy} = \frac{1}{2}\left(\frac{\text{TP}}{\text{TP}+\text{FN}} + \frac{\text{TN}}{\text{TN}+\text{FP}}\right) \tag{2}$$

$$\text{Precision} = \left(\frac{\text{TP}}{\text{TP}+\text{FP}}\right) \tag{3}$$

$$\text{Recall} = \left(\frac{\text{TP}}{\text{FP}+\text{FN}}\right) \tag{4}$$

$$\text{MCC} = \frac{\text{TP}\times\text{TN} - \text{FP}\times\text{FN}}{\sqrt{(\text{TP}+\text{FP})(\text{TP}+\text{FN})(\text{TN}+\text{FP})(\text{TN}+\text{FN})}} \tag{5}$$

where TP denotes true positives, TN denotes true negatives, FP denotes false positives, and FN denotes false negatives.

3) Hyperparameter Tuning: After selecting the final model, hyperparameters were optimized using the Random Search method, coupled with fivefold cross-validation, with the aim of maximizing the MCC coefficient. In the validation phase, a reduced number of folds (fivefold) was adopted compared to training to enable efficient optimization in high computational cost environments. The process included 100 iterations and 50 hyperparameter combinations to ensure randomness in the search, using the default functions provided by PyCaret to identify configurations that balanced model complexity and reduced overfitting [23], [66]. Although tuning the hyperparameters exclusively for the best-performing model may favor those that already stand out in their baseline configuration, the high computational cost associated with processing a

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TABLE IV
COMPARISON OF ML MODEL PERFORMANCE USING DIFFERENT FEATURE SETS DURING TRAINING

Model	Partition	Accuracy			MCC			Precision			Recall			Processing Time (s)		
		RGB	MSI	RGB MSI	RGB	MSI	RGB MSI	RGB	MSI	RGB MSI	RGB	MSI	RGB MSI	RGB	MSI	RGB MSI
XGBoost	Train	0.71	0.78	0.88	0.42	0.57	0.76	0.47	0.60	0.78	0.45	0.58	0.77	8.79	5.53	13.90
	Test	0.70	0.73	0.83	0.39	0.47	0.66	0.43	0.51	0.68	0.43	0.50	0.67	—	—	—
CatBoost	Train	0.71	0.77	0.88	0.42	0.54	0.76	0.47	0.57	0.78	0.45	0.56	0.77	19.21	11.95	22.74
	Test	0.70	0.73	0.83	0.39	0.47	0.66	0.43	0.50	0.68	0.43	0.49	0.68	—	—	—
LightGBM	Train	0.71	0.78	0.87	0.41	0.56	0.74	0.46	0.59	0.75	0.44	0.58	0.75	23.50	16.44	24.09
	Test	0.70	0.73	0.82	0.39	0.47	0.65	0.43	0.50	0.67	0.43	0.49	0.67	—	—	—
RF	Train	0.83	0.93	0.96	0.66	0.85	0.93	0.68	0.87	0.94	0.67	0.86	0.93	23.20	9.26	44.84
	Test	0.68	0.74	0.83	0.36	0.48	0.67	0.39	0.51	0.69	0.40	0.50	0.68	—	—	—
ETC	Train	0.83	0.92	0.96	0.66	0.85	0.93	0.69	0.86	0.94	0.67	0.86	0.93	14.94	4.04	21.13
	Test	0.68	0.73	0.83	0.35	0.47	0.66	0.39	0.50	0.69	0.39	0.49	0.68	—	—	—
LR	Train	0.68	0.72	0.80	0.36	0.44	0.61	0.42	0.48	0.64	0.40	0.47	0.63	79.47	23.58	121.17
	Test	0.69	0.72	0.80	0.37	0.43	0.60	0.42	0.47	0.63	0.41	0.46	0.62	—	—	—
KNN	Train	0.74	0.78	0.89	0.48	0.57	0.79	0.52	0.60	0.81	0.51	0.59	0.80	17.34	2.80	27.40
	Test	0.68	0.71	0.81	0.35	0.42	0.62	0.38	0.46	0.64	0.39	0.45	0.63	—	—	—
DT	Train	0.82	0.91	0.95	0.65	0.83	0.90	0.68	0.84	0.90	0.66	0.84	0.90	9.54	5.30	37.32
	Test	0.67	0.70	0.79	0.33	0.40	0.57	0.37	0.43	0.60	0.37	0.43	0.60	—	—	—
LDA	Train	0.67	0.70	0.77	0.34	0.40	0.55	0.38	0.43	0.58	0.38	0.43	0.57	1.42	0.40	2.74
	Test	0.68	0.70	0.77	0.35	0.39	0.55	0.39	0.43	0.58	0.39	0.43	0.57	—	—	—
QDA	Train	0.67	0.71	0.77	0.32	0.41	0.52	0.40	0.46	0.58	0.37	0.46	0.57	1.43	0.41	2.68
	Test	0.67	0.71	0.77	0.32	0.40	0.52	0.40	0.45	0.57	0.38	0.46	0.57	—	—	—
SVM-L	Train	0.66	0.67	0.76	0.32	0.34	0.52	0.37	0.43	0.56	0.36	0.38	0.54	2.51	0.83	4.81
	Test	0.67	0.67	0.76	0.33	0.33	0.52	0.38	0.41	0.57	0.37	0.38	0.55	—	—	—
Ridge	Train	0.65	0.66	0.72	0.36	0.35	0.44	0.39	0.45	0.58	0.33	0.35	0.47	1.21	0.38	1.83
	Test	0.65	0.66	0.72	0.36	0.36	0.46	0.37	0.47	0.56	0.33	0.35	0.46	—	—	—
NB	Train	0.67	0.68	0.73	0.32	0.34	0.43	0.36	0.37	0.46	0.37	0.40	0.49	1.45	0.46	2.35
	Test	0.67	0.68	0.73	0.33	0.33	0.43	0.37	0.37	0.46	0.38	0.40	0.49	—	—	—
AdaBoost	Train	0.53	0.56	0.60	0.10	0.14	0.23	0.22	0.24	0.30	0.12	0.17	0.23	34.53	18.80	107.55
	Test	0.53	0.56	0.59	0.09	0.14	0.23	0.16	0.23	0.32	0.12	0.18	0.23	—	—	—

high-dimensional dataset led to focusing efforts on the models that delivered the most promising results.

F. Blending

In addition to the individual evaluation of models, a model combination approach known as blending was adopted to enhance predictive performance. The blending process was carried out by selecting the top-performing models and combining their predictions in a weighted manner. Initially, five models were considered, selecting the five best-performing models based on evaluation metrics using two selection strategies: 1) ranking models according to their overall performance regardless of the algorithms used; and 2) avoiding models based on the same underlying method, such as XGBoost, LightGBM, and CatBoost, which are all gradient boosting methods. Subsequently, a soft voting strategy was applied to consolidate the predictions by averaging the predicted probabilities. A summary of the main steps of the process is presented in Fig. 2.

III. RESULTS

A. Evaluation of Predictive Performance Across Different Feature Sets

XGBoost stood out with the highest overall scores for accuracy, MCC, precision, and recall across all configurations (i.e., RGB, MSI, and RGB + MSI) in both the training and test sets, although its performance was closely followed by other models (Table IV), such as CatBoost, LightGBM, RF, and ETC. The highest values were observed in the RGB + MSI partition, including the MCC scores, where RF led this metric (training: 0.93; test: 0.67), followed by ETC (training: 0.93; test: 0.66), XGBoost (training: 0.76; test: 0.66), and CatBoost (training: 0.76; test: 0.66). On the other hand, although the Ridge, NB, QDA, LDA, Ridge, and SVM-L models had lower execution times (1.83, 2.35, 2.68, 2.74, and 4.81 s, respectively), they showed inferior performance metrics. Conversely, the LR and AdaBoost models had the highest processing times (121.17 and 107.55 s, respectively), although LR showed performance levels close to those of the top models.

When analyzing the performance of the algorithms across different classes, we observed significant variations in the levels of result dispersion, along with lower performance on the training data (Fig. 3). These differences become particularly evident when using more robust metrics for multiclass data, such as the MCC index (Fig. 3). For example, although the ETC and RF models showed high average MCC values, the results on the test data revealed greater dispersion, while dispersion on the training data was minimal, suggesting the occurrence of overfitting to the training set. The XGBoost, CatBoost, LightGBM, and KNN models, which also stood out for their high average scores, showed greater consistency between training and test data, indicating better generalization. The LR model demonstrated even greater stability across different partitions, with nearly identical distributions of its results. LDA, QDA, SVM-L, and NB showed good generalization despite their lower performance levels. In contrast, the DT model not only exhibited greater dispersion in test data but also had lower average values. The remaining models not shown in Fig. 3 were excluded from evaluation because they

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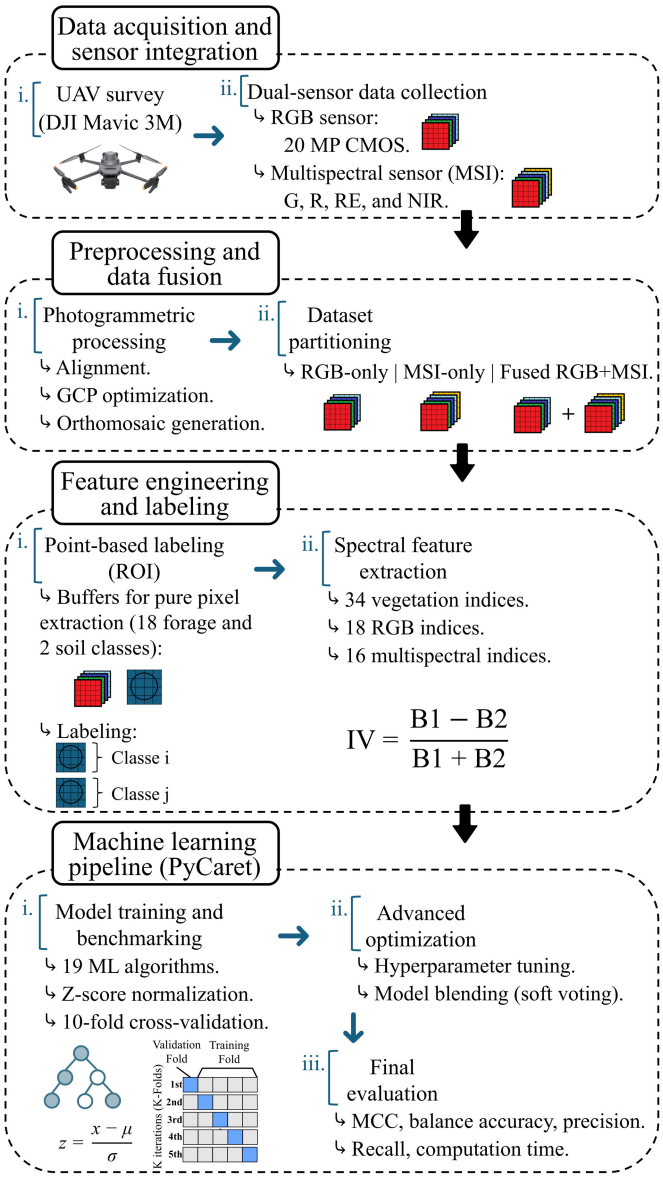


Fig. 2. Summary of the crop classification process. Methodological pipeline for UAV-based forage classification using multispectral imagery and ML. Data acquisition and sensor integration: aerial surveys were conducted with a DJI Mavic 3M UAV equipped with a dual-sensor system comprising a 20-MP RGB camera and a multispectral sensor capturing green, red, red-edge, and near-infrared bands. Preprocessing and data fusion: raw imagery underwent photogrammetric processing including image alignment, GCP optimization, and orthomosaic generation, followed by dataset partitioning into three configurations: RGB-only, MSI-only, and fused RGB + MSI. Feature engineering and labeling: ROIs were defined using point-based labeling with buffered pure-pixel extraction across 18 forage and 2 soil classes; spectral feature extraction yielded 68 indices in total, comprising 34 vegetation, 18 RGB, and 16 multispectral indices, computed via a normalized difference formula. ML pipeline: model training and benchmarking were performed in PyCaret using 19 ML algorithms with Z-score normalization and tenfold cross-validation, followed by advanced optimization through hyperparameter tuning and soft-voting model blending; final model evaluation was based on MCC, balanced accuracy, precision, recall, and computation time.

were unable to predict all available classes. As observed in the evaluation presented in Table IV, the combination of RGB + MSI features was decisive for achieving the best performance and was therefore selected for the subsequent analyses.

B. Evaluation of Predictive Performance by Class on the Test Data Using the Best Feature Set

The XGBoost, CatBoost, RF, and ETC models consistently stood out, achieving MCC coefficients above 0.75 in 39% of the classes (seven classes), indicating high agreement between predictions and observed values, as well as superior predictive performance in multiclass problems. LightGBM followed closely, with 33% of the classes (six classes) reaching the same threshold (Fig. 4). In contrast, the LR, KNN, DT, QDA, LDA, and SVM-L models showed lower consistency, with only 28% of the classes (five classes) attaining MCC values above 0.75. The NB model performed worst, with only 11% of the classes (two classes) exceeding this threshold. Indeed, most classes showed high MCC coefficients across nearly all evaluated models, particularly the classes pigeon pea, maize, agave, soil, and mulching. On the other hand, the classes signal grass, Mombaça, Tanzania, IPA, MIU, and Prickly pear had lower coefficients, highlighting specific classification challenges in these categories. Performance on the training sets was generally higher than on the test sets, which is expected due to the direct fitting to training data.

Although RF achieved slightly better overall metrics (accuracy, MCC, precision, and recall) using RGB and multispectral data, XGBoost delivered very similar results in both training and testing, demonstrating robust predictive capacity and superior performance when considering all data sources used for modeling (RGB, multispectral, and RGB + multispectral). Moreover, XGBoost stood out for its shorter processing time among the high-performing models (13.90 s), being significantly faster than RF and other well-performing models (Fig. 4). This processing efficiency, combined with a strong balance between performance and generalization, was decisive in selecting XGBoost as the best model, particularly in scenarios requiring high accuracy with lower computational cost. As a result, both RF and XGBoost were selected for hyperparameter tuning.

C. Evaluation of the Predictive Performance of Tuned and Combined Models

For the construction of the combined model, the following combinations were used in the list.

1) XGBoost, RF, LR, KNN, and DT, in the order presented in Table IV, while avoiding models with similar structures, such as CatBoost and LightGBM, both of which are based on gradient boosting theory, like XGBoost. Although ETC, despite having an even more randomized tree structure than RF, produced results similar to RF in this study.
2) Following the same order from the table without considering model similarity: XGBoost, CatBoost, LightGBM, RF, and ETC. The models were evaluated together with the tuned RF and XGBoost models.

In general, it was observed that the models exhibited high values in the training set, with scores above 0.80 in most cases, while a significant reduction was noted in the testing set, indicating possible overfitting (Fig. 5). Although improvements were observed in the training data, especially

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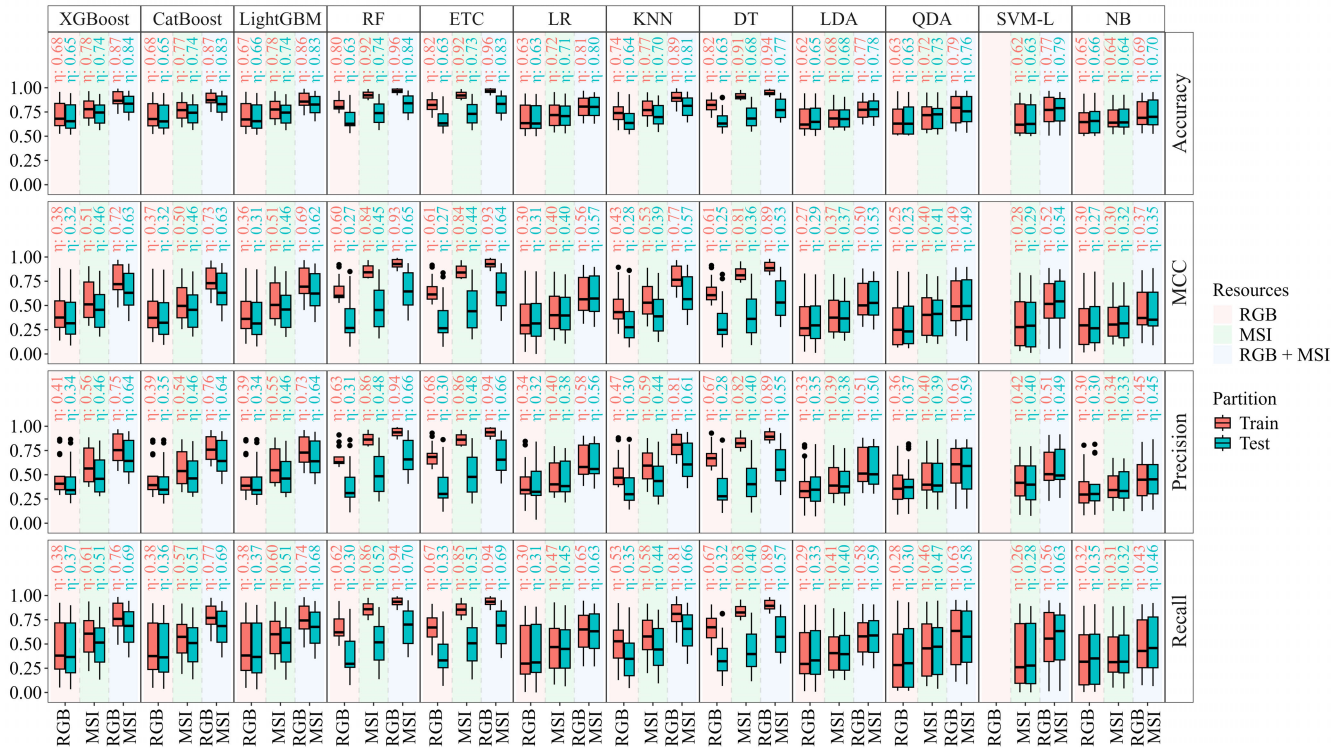


Fig. 3. Predictive performance of ML models across different datasets. MCC: Matthews correlation coefficient; RGB: red, green, and blue; and MSI: multispectral. The value shown at the top of the boxplot indicates the median η.

when compared to XGBoost, the blended models did not result in improvements in the testing data, with the Blend Select Models showing the worst overall performance. Similarly, the tuned models did not demonstrate improvements over their original versions and, in some cases, even worsened the results presented.

D. Predictive Performance Evaluation by Class for the Tuned and Combined Data

When evaluating the models' performance based on the MCC index by class [Fig. 6(a) and (b)], it was observed that the tuned and combined models did not show improvements in any of the classes, particularly in those with greater classification difficulty, such as the forage cactus clones (IPA, OEM, and MIU) and the grasses Mombaça and Tanzania. Even for soil and dead mulch, which were easier to predict, these models demonstrated reduced performance. Thus, XGBoost remained the best-performing model evaluated.

The analysis of the XGBoost model's performance in vegetation classification using the confusion matrix revealed high accuracy in identifying certain classes, while others exhibited greater confusion. The distribution of the model's predictions for each observed class highlights a predominance of correct classifications for more distinct categories, such as Agave, Pigeon pea, Gliricidia, Leucaena, Sorghum, Maize, soil, and dead mulch. However, considerable dispersion of predictions was observed for classes such as Tanzania, Mombaça, and the cacti clones (IPA, MIU, OEM, and prickly pear cactus) (Fig. 7). The normalized confusion matrix [Fig. 7(b)] reinforces this trend, indicating that the error rates, although higher, were concentrated in the confusion among classes from similar groups. An aggregation of errors (dashed lines, Fig. 7) can be noted among classes belonging to the C3, C4, CAM, and nonvegetation groups.

E. Feature Importance Analysis

The mean absolute SHAP values obtained for the XGBoost model trained with the RGB + MSI dataset are presented in Fig. 8. The ranking indicates that a small subset of variables concentrated the largest contributions to the model predictions. ExB (0.79) and ExG (0.72) presented the highest importance values, followed by HUE (0.58), the raw green band from the RGB sensor (0.53), and MTCI (0.49). Other variables with relatively high contributions included GOSAVI (0.45), GNDVI (0.39), SFDVI (0.37), NDVI (0.35), and the raw red band from the RGB sensor (0.34). Regarding sensor origin, four of the five highest-ranked variables were derived from RGB imagery, whereas MSI-based features were more frequent in the intermediate portion of the ranking, particularly indices sensitive to near-infrared reflectance and red-edge transitions (e.g., MTCI, GOSAVI, GNDVI, and NDRE). In contrast, some ratio-based indices presented minimal contributions, including GRRI, GCC, RVI (0.005), and GRRI_{M}.

IV. Discussion

A. Benefits of Sensor Data Fusion

Merging RGB and multispectral imagery enhanced discriminative capacity compared to single-sensor datasets. Similar findings were reported by [22] for grain crops and by [67] for

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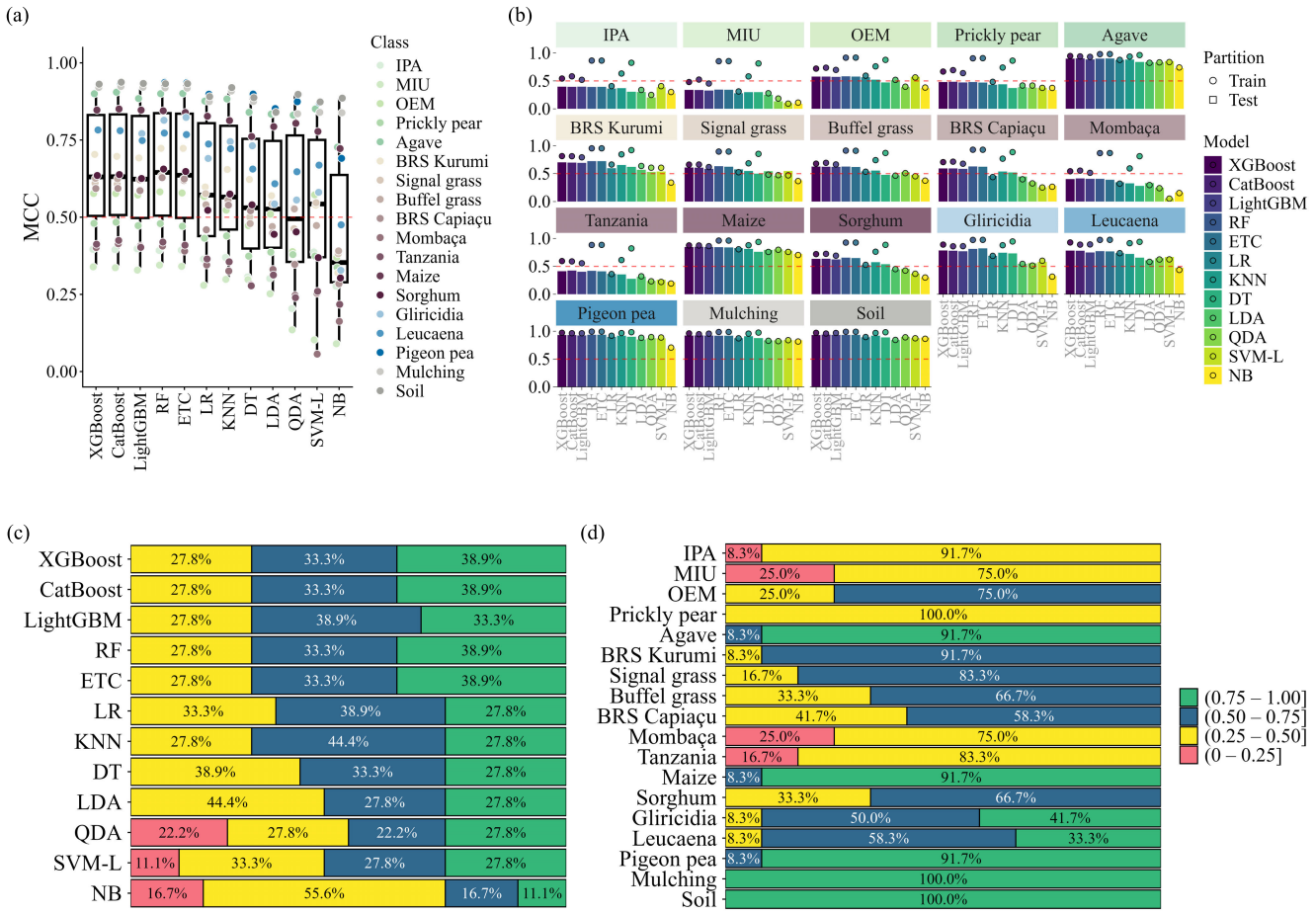


Fig. 4. Multimodel performance analysis and class-specific predictability using fused RGB + multispectral data. (a) Distribution of MCC across 15 ML algorithms; boxplots represent the variability of performance across all 18 classes, with the red dashed line indicating the 0.5 performance threshold. (b) Individual model performance per class; bars represent the test partition and points represent the train partition, highlighting model stability; facet backgrounds are color-coded by functional group: CAM species (green), C4 grasses (brown/purple), C3 legumes (blue), and nonvegetation (gray). (c) Relative frequency of MCC performance intervals for each algorithm, ranking models by their ability to achieve high-consistency classification (MCC > 0.75). (d) Predictability profile per class across all tested models, showing the proportion of algorithms that reached specific MCC intervals, identifying the most robustly classified species.

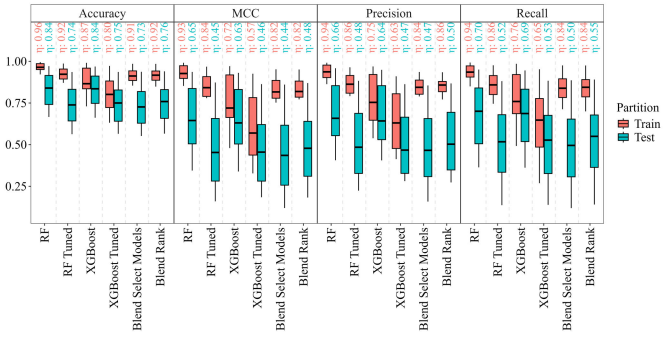


Fig. 5. Performance evaluation of the selected, tuned, and combined models on training and testing data. Blend select models: XGBoost, RF, LR, KNN, and DT. Blend rank: XGBoost, CatBoost, LightGBM, RF, and ETC. MCC: Matthews correlation coefficient. The value shown at the top of the boxplot indicates the median η.

tree species classification. More recently, Fu et al. [68] demonstrated accuracy gains of up to 41.60% in canopy chlorophyll retrieval through spectral complementarity, further validating the benefits of multisensor integration. This enhancement is likely due to the increased feature space available to identify optimal splits for class separation [22]. In this study, the balanced presence of both RGB and MSI features among the top-ranked predictors (Fig. 8) justifies the marginal gains observed in the fused dataset. The high ranking of RGB-derived indices, such as ExG and ExB, is consistent with their ability to maximize the contrast between green biomass and nonphotosynthetic backgrounds (soil and mulching) [57]. Concurrently, the significance of MSI-specific indices, such as MTCI and GOSAVI, underscores the importance of the red-edge and NIR regions for characterizing forage crops. The MTCI is widely recognized for its sensitivity to chlorophyll content while being less affected by atmospheric or soil interference [50], and GOSAVI is optimized for soil-adjusted vegetation monitoring in open-canopy conditions [54].

Nevertheless, increasing dimensionality does not always guarantee performance gains and can increase computational demands [69]. For example, while the RF algorithm demonstrated competitive accuracy, it incurred higher computational

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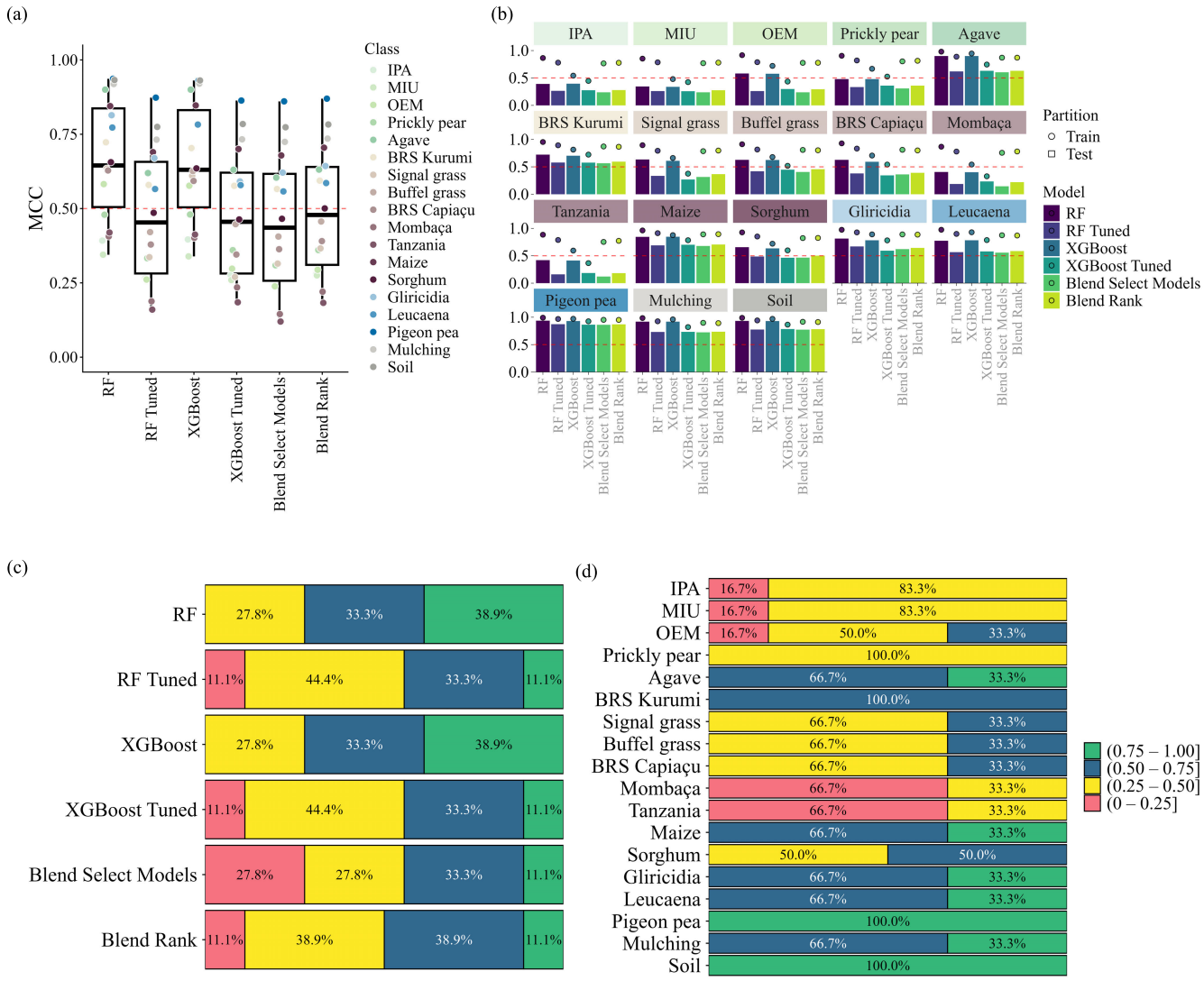


Fig. 6. Performance assessment of hyperparameter tuning and ensemble strategies (blends) on class-specific classification. (a) Distribution of MCC for base models, tuned versions, and ensemble blends; boxplots illustrate the performance spread across the 18 classes, with the red dashed line at 0.5 indicating the moderate performance threshold. (b) Comparison of training (dots) and testing (bars) performance per class for each modeling strategy; facet backgrounds are color-coded by functional group: CAM species (green), C4 grasses (brown/purple), C3 legumes (blue), and nonvegetation (gray). (c) Relative frequency of MCC performance intervals per modeling approach, highlighting the tradeoffs between default configurations, tuned models, and ensemble methods. (d) Predictability profile per class across all tested models, showing the proportion of algorithms that reached specific MCC intervals, identifying the most robustly classified species.

costs compared to XGBoost, highlighting the importance of feature selection to retain only the most impactful variables for model efficiency [23].

B. Model Performance

XGBoost demonstrated an optimal balance between predictive performance and computational efficiency, making it suitable for robust, high-speed applications. Its training-to-test stability suggests a lower propensity for overfitting. Recurring challenge in ML under high data variability [70], [71]. While RF and ETC showed competitive accuracy, their higher result dispersion suggests XGBoost is a more reliable alternative for heterogeneous environments. These findings are consistent with previous studies that employed XGBoost among models designed for classification tasks [28], [72], [73].

Such performance stems from a learning strategy that efficiently adapts to the structure of the analyzed data [74]. While RF and ETC employ bagging to reduce variance through averaging across independent trees, XGBoost adopts a sequential gradient boosting approach based on the minimization of a regularized objective function [75]. The availability of multiple regularization parameters, including explicit L1 and L2 terms, contributes to mitigating overfitting in highly dimensional and correlated feature spaces [75]. Consequently, the advantage of XGBoost reflects its ability to balance model complexity and generalization in heterogeneous agricultural environments. Although gradient boosting algorithms such as LightGBM and

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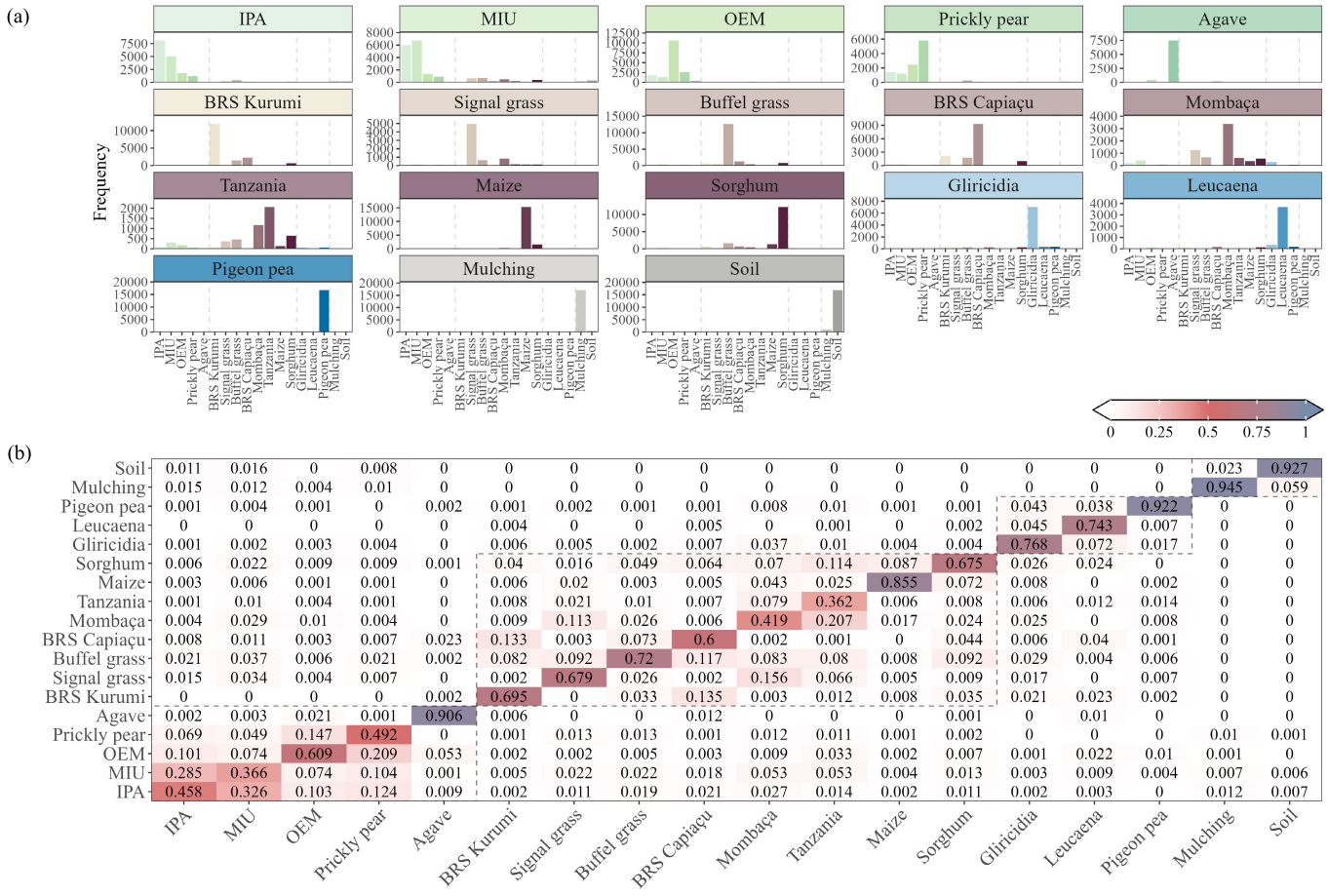


Fig. 7. Performance of the XGBoost model in forage species classification using fused RGB + MSI imagery. (a) Frequency distribution of model predictions for each observed class, organized by functional group: CAM species (green), C4 grasses (brown/purple), C3 legumes (blue), and nonvegetation classes (gray); dashed vertical lines delimit transitions between functional groups. (b) Normalized confusion matrix showing the proportion of predictions per observed class; values along the diagonal represent per-class accuracy, while off-diagonal values indicate misclassification rates; dashed lines delimit functional group boundaries, and color intensity reflects prediction proportion from low (white) to intermediate (red) to high (blue).

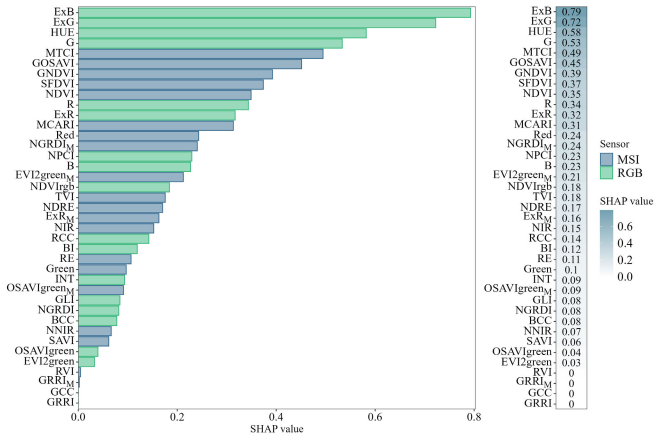


Fig. 8. Mean absolute SHAP values obtained for the XGBoost model trained with the RGB + MSI (multispectral) dataset. R, G, and B represent the red, green, and blue colors of the RGB sensor.

CatBoost share similar conceptual foundations, their inferior performance in this study may be related to the interaction between data structure and the specific tree construction and optimization strategies adopted by each method [74]. Finally, the generalization observed for LR, despite its lower absolute accuracy, raises questions regarding the tradeoff between interpretability and performance in applications that require transparency in decision-making, given that the estimation process of XGBoost is often regarded as a black box [76]. Still, the technical advantages of its architecture provide a compelling justification for its selection.

C. Effects of Hyperparameter Tuning and Model Blending

Although hyperparameter tuning and model blending have improved classification accuracy in various ML applications, in this study, both practices failed to yield improvements and, in some cases, even worsened the results. Regarding hyperparameters, depending on the model, the possible combinations can result in a large number of possibilities [77]. Moreover, depending on the strategy employed, it may not be possible to find a combination that improves the outcomes unless the entire search space is explored, which can render the process highly burdensome [78], [79]. Therefore, it cannot be stated that the tuning process is not useful in the studied case.

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Concerning the use of blending models, the high confusion among certain classes may have resulted in models that did not provide additional contributions to the ensemble, as evidenced by the performance reduction when compared solely to XGBoost. The stagnation of the class-wise MCC, even in simpler categories, demonstrates that conventional strategies of model blending and tuning may be insufficient in contexts with high interclass similarity, reinforcing the need for approaches that prioritize information extraction from the data [80].

The use of PyCaret offered significant advantages for this study, particularly during the initial phase of model exploration. Its ability to automate comparisons among different algorithms, configure standardized preprocessing steps, and rapidly generate evaluation metrics provided a comprehensive overview of model behavior and served as a consistent starting point for initial selection [34], [81]. The environment also supports fine-tuning of models, including the specification of customized hyperparameters and the adjustment of search ranges during optimization. While AutoML is effective for providing an initial screening and accelerating development, our results indicate that applications in complex agricultural scenarios benefit from subsequent manual customization steps to maximize predictive accuracy and robustness [82], [83].

D. Classification Challenges Among Phenotypically Similar Crops

The high accuracy achieved for more distinct categories suggests that the XGBoost model effectively captures structural and spectral patterns of these classes. However, the confusion observed among more similar groups highlights limitations in the model's discriminative power, as evidenced by concentrated errors among classes from closely related phenotypic groups (C3, C4, CAM, and nonvegetation). This suggests that the model may require more robust differentiation between these groups, as well as accounting for variations in lighting conditions, soil backgrounds, and mixed pixels [84]. Reflecting the need for enhancements in the model's approach, we believe that adjustments in data preprocessing or the incorporation of additional features could improve the accuracy of discrimination between these classes. In other studies, Hoppe et al. [72] also identified difficulties in classifying more similar species, such as wheat and rye, leading to increased confusion in the models. Meanwhile, Blickensd örfer et al. [1] found that although classification among taxonomically similar groups was feasible, noticeable confusion persisted within those groups. Similarly, Deng et al. [2] reported challenges in differentiating luffa and sweet potato due to similarities in leaf color, texture, and shape. In turn, Fan et al. [5] suggested that classifying crop groups rather than individual crop types would be more appropriate when using freely available satellite imagery, recommending higher resolution satellites for crop type classification. Nevertheless, we found that even at higher resolutions, distinguishing between similar crop types remains a challenge.

Although DL-based approaches are often prioritized for complex classification tasks [35], [36], recent evidence indicates that classical ML methods may exhibit superior performance in specific agricultural scenarios [85], [86]. Wang et al. [85] demonstrated that, even when integrating multiple sensors (LiDAR, thermal, MSI, and RGB), CNN models underperformed relative to RF in small-scale areas with limited training samples. Complementarily, Zhao et al. [86] showed that the explicit incorporation of spectral and spatial information through the spectral–spatial conditional random fields (SCRFs) algorithm outperformed conventional approaches such as support vector machines (SVMs) and RF, as well as methods including spectral–spatial residual network (SSRN) and multiresolution segmentation in eCognition, by strengthening the separability among heterogeneous, even spectrally similar, crops. While Jia [87] reported substantially limited performance in maize classification (Recall = 0.17) using Sentinel-2 data combined with a CNN–RNN model, the present study demonstrates that UAV data associated with the XGBoost algorithm achieve significantly higher reliability for maize.

In the present study, the highest degree of confusion occurred in the classification of IPA, MIU, and prickly pear cactus, primarily due to structural similarities among these crops, which are characterized by complex structural dimensions, with cladodes frequently oriented vertically, thereby limiting the area captured by UAV-mounted sensors. Nevertheless, the concentration of model errors among closely related classes can be interpreted as a positive outcome, ensuring greater accuracy in more generalized classifications and highlighting these specific challenges [5]. Our results indicate that the quality of the input data and the definition of phenotypic groups are critical factors for the model's success. In agreement with evidence previously reported in the literature, reinforce the need for future investigations to incorporate additional sources of information, such as structural attributes, textural metrics, and alternative color spaces, as suggested by the limitations observed in both DL models and classical ML algorithms applied to highly heterogeneous agricultural environments [23], [85], [86], [87].

E. Study Limitations and Future Perspectives

Although this study provides relevant contributions to vegetation classification using UAV and ML, some limitations must be considered. First, image quality remains susceptible to variations in lighting, reflections, and atmospheric fluctuations, which can bias spectral data and introduce noise into the classification models, thereby affecting result accuracy. Furthermore, while RGB and multispectral fusion provided high accuracy, the spectral overlap observed between certain C4 grasses and cacti clones suggests that future extensions to hyperspectral imagery could provide the narrowband information necessary to resolve these fine-scale phenological differences.

In addition, the exclusive use of PyCaret, although facilitating automated modeling, restricts the customization of algorithms and optimization strategies. In this study, we prioritized pixel-based ML to establish a computationally efficient and accessible baseline for real-time forage monitoring; however, we recognize that the exclusion of DL and object-based image analysis (OBIA) is a methodological boundary. While

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pixel-based approaches are highly effective for ultrahigh-resolution mapping with limited training samples, DL and OBIA frameworks could potentially offer further insights into the complex spatial patterns and structural geometries of forage crops in future investigations.

In particular, hyperparameter optimization was applied only to the best-performing models in the initial screening (RF and XGBoost), an approach that may have favored algorithms performing well with default settings to the detriment of models such as SVM, which are highly sensitive to tuning. The high computational cost of exhaustively optimizing all 20 models rendered this approach impractical. Finally, the exclusion of agronomic variables (e.g., irrigation levels, fertilization doses) prevents the evaluation of their confounding effects on spectral responses. Future research should therefore adopt methodologies that allow for hyperspectral sensing, DL-based frameworks, and careful variable selection, and algorithm customization to explore more robust configurations, as well as conduct broader hyperparameter optimization prior to final model selection to assess whether other algorithms could outperform those selected here, in addition to detailed analyses of the effects of in-field variables.

V. Conclusion

This study demonstrates that the strategic integration of ultrahigh-resolution RGB and multispectral data, processed through optimized ML workflows, provides a robust framework for mapping complex and heterogeneous forage systems. The fusion of these sensors, combined with the XGBoost algorithm, proved to be the most effective approach, offering a superior balance between predictive accuracy, model generalization, and computational efficiency compared to 18 other evaluated models.

Despite the high performance achieved, some limitations must be acknowledged. Image quality remained susceptible to variations in illumination and atmospheric conditions, which can introduce noise into the spectral data. The classification of phenotypically similar classes, such as certain cacti clones and C4 grasses, remains a challenge due to spectral overlap and vertical canopy structures that limit sensor visibility. Furthermore, while the AutoML-based approach facilitated extensive benchmarking, it may overlook specific algorithmic sensitivities that exhaustive manual tuning could reveal.

The findings underscore that while advanced ensemble methods like XGBoost are highly effective, the quality and dimensionality of the input data remain the primary drivers of success in precision livestock farming applications. Future research should focus on incorporating structural attributes from 3-D point clouds, exploring DL architectures for hierarchical feature extraction, and evaluating the confounding effects of agronomic variables on spectral responses to further enhance classification reliability in diverse agricultural scenarios.

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